Homework 10

Nov 19, 2025

1 (30'). Use first-order Peano arithmetic formulas to express "$m = 2^n$".

2 (30'). Prove P is closed under Kleene star. In other words, if a language L belongs to P , then so does L^*.

3 (40'). Let

$$L_1 = \{(\alpha, x) : \text{TM } M_\alpha \text{ accepts input } x\},$$

and

$$L_2 = \{\alpha : \text{TM } M_\alpha \text{ halts on the empty input}\}.$$

Prove that $L_1 \leq_T L_2$.